

SIMATS ENGINEERING

ASSESSMENT TOOL 4

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

Time Duration: 1 Hour

QUESTIONS: 5

Total Marks:30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I BAIRAGANI DINESH 192372189 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **BAIRAGANI DINESH 192372189**

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ($\square$) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
 2. Convert all implications into CNF.
 3. Negate the goal.
 4. Apply the Resolution Algorithm to derive the Empty Clause ($\square$).
 5. Explain each resolution step and write the final conclusion.
- 1.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

		requirements accurately			
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Question 1 – Rain and Wet Ground

1. Propositional Logic Representation

Let:

- R : It is raining.
- W : The ground is wet.

Premises:

1. If it rains, then the ground becomes wet: $R \rightarrow W$
2. It is raining: R

Goal:

- W (The ground is wet)

2. Conversion to CNF (Conjunctive Normal Form)

- **Premise 1:** $R \rightarrow W \equiv \neg R \vee W$ (Clause 1)
- **Premise 2:** R (Clause 2)

3. Negation of the Goal

- **Negated Goal:** $\neg W$ (Clause 3)

4. Resolution Algorithm Step-by-Step

- **Clause 1:** $\neg R \vee W$
- **Clause 2:** R
- **Clause 3:** $\neg W$

Steps:

1. Resolve **Clause 1** ($\neg R \vee W$) and **Clause 2** (R):
 - Eliminating R and $\neg R$ yields **Clause 4:** W
2. Resolve **Clause 4** (W) and **Clause 3** ($\neg W$):
 - Eliminating W and $\neg W$ yields the **Empty Clause** ($\square$).

5. Conclusion

Since the empty clause ($\square$) is derived, a contradiction is reached. Thus, the goal is proved: **The ground is wet.**

Question 2 – Student Assignment Submission

1. Propositional Logic Expressions

Let:

- S : Rahul submits the assignment.
- M : Rahul receives internal marks.

Premises:

1. If a student submits the assignment, then the student receives internal marks:
 $S \rightarrow$
2. Rahul submitted the assignment: S

Goal:

- M (Rahul receives internal marks)

2. Conversion to CNF

- **Clause 1:** $\neg S \vee M$
- **Clause 2:** S

3. Negated Goal

- **Clause 3:** $\neg M$

4. Resolution Steps

1. Resolve **Clause 1** ($\neg S \vee M$) and **Clause 2** (S):
 - Yields **Clause 4:** M
2. Resolve **Clause 4** (M) and **Clause 3** ($\neg M$):
 - Yields the **Empty Clause** ($\square$).

5. Conclusion

The derivation of the empty clause ($\square$) proves that **Rahul receives internal marks**.

Question 3 – Library Membership

1. Propositional Logic Representation

Let:

- L : Priya is a library member
- B : Priya can borrow books.

Premises

1. If a person is a library member, then the person can borrow books: $L \rightarrow B$
2. Priya is a library member: L

Goal:

- B (Priya can borrow books)

2. Conversion to CNF

- **Clause 1:** $\neg L \vee B$
- **Clause 2:** L

3. Negated Goal

- **Clause 3:** $\neg B$

4. Resolution Execution

1. Resolve **Clause 1** ($\neg L \vee B$) and **Clause 2** (L):
 - Result: **Clause 4:** B
2. Resolve **Clause 4** (B) and **Clause 3** ($\neg B$):
 - Result: **Empty Clause** ($\square$)

5. Final Conclusion

The Resolution Algorithm derives an empty clause ($\square$), successfully proving that **Priya can borrow books**.

Question 4 – Placement Eligibility

1. Propositional Logic Statements

Let

- A : Arun cleared the aptitude test.
- P : Arun is eligible for placement.

Premises:

1. $A \rightarrow P$
2. A

Goal:

- P (Arun is eligible for placement)

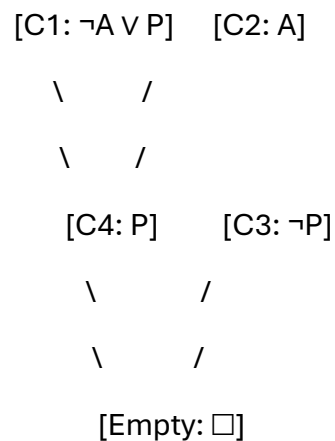
2. CNF Clauses

- **Clause 1 (C_1):** $\neg A \vee P$
- **Clause 2 (C_2):** A

3. Negated Goal

- **Clause 3 (C_3):** $\neg P$

4. Resolution Algorithm & Resolution Tree



Step-by-Step Resolution:

1. From $C_1 (\neg A \vee P)$ and $C_2 (A)$, resolve on A to obtain $C_4 (P)$.
2. From $C_4 (P)$ and $C_3 (\neg P)$, resolve on P to obtain the **Empty Clause ($\square$)**.

5. Conclusion

The resolution tree leads directly to the empty clause ($\square$), establishing that **Arun is eligible for placement**.

Question 5 – Access Control System

1. Propositional Logic Representation

Let:

- P : User enters the correct password
- A : User is authenticated.
- G : User is granted access.

Premises:

1. If a user enters the correct password, then the user is authenticated: $P \rightarrow A$
2. If a user is authenticated, then the user is granted access: $A \rightarrow G$
3. The user entered the correct password: P

Goal:

- G (User is granted access)

2. Conversion to CNF Clauses

- **Premise 1:** $P \rightarrow A \equiv \neg P \vee A$ (**Clause 1**)
- **Premise 2:** $A \rightarrow G \equiv \neg A \vee G$ (**Clause 2**)
- **Premise 3:** P (**Clause 3**)

3. Negation of Goal

- **Negated Goal:** $\neg G$ (**Clause 4**)

4. Resolution Algorithm to Derive Empty Clause ($\square$)

- **Clause 1:** $\neg P \vee A$
- **Clause 2:** $\neg A \vee G$
- **Clause 3:** P
- **Clause 4:** $\neg G$

Resolution Steps Explained:

1. **Step 1:** Resolve **Clause 1** ($\neg P \vee A$) with **Clause 3** (P).
 - *Explanation:* The literal P in Clause 3 cancels out $\neg P$ in Clause 1
 - *Derived Clause 5:* A
2. **Step 2:** Resolve **Clause 2** ($\neg A \vee G$) with **Clause 5** (A).
 - *Explanation:* The literal A in Clause 5 cancels out $\neg A$ in Clause 2
 - *Derived Clause 6:* G

3. **Step 3:** Resolve **Clause 6** (G) with **Clause 4** ($\neg G$)

- *Explanation:* The literal G in Clause 6 cancels out $\neg G$ in Clause 4.
- *Derived Clause:* **Empty Clause** ($\square$)

5. Final Conclusion

Through proof by contradiction, deriving the **Empty Clause** ($\square$) proves that the original hypothesis (that access is denied) cannot hold. Therefore, **the user is granted access.**